

Answer Key — Biology Grade 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals • Lesson 7: Putting It All Together: Final Explanation of How Animals Keep a Balanced Internal Environment and Remove Toxic Wastes

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
D	D	C	B	D	C	D	D	D

USE AFTER: PREDICT PHASE — Predict: goat deprived of water for a day in Kajiado (Think-Pair-Share prediction)

Q1. A goat in Kajiado is kept without water for a whole day on a hot afternoon. What is the most likely result?

- A. Its urine becomes more dilute because the kidney stops working to save energy
- B. Its urine stays the same because goats only control water by drinking less
- C. Its blood water level rises because the body makes extra water from urine
- D. Its urine becomes more concentrated because ADH makes the kidney reabsorb more water**

Correct answer: D. Water loss raises the concentration of the blood, so ADH is released and the kidney tubules reabsorb more water. This gives a small volume of concentrated urine and helps keep the blood water level steady.

USE AFTER: OBSERVE PHASE — Observe: kidney specimen Predict-Observe-Explain check (cortex, medulla, pelvis, nephron)

Q2. Learners cut open a goat kidney and review how a nephron works. Which statement is correct?

- A. The loop of Henle collects urine and carries it to the bladder
- B. The glomerulus reabsorbs water and the tubule filters the blood
- C. The pelvis filters the blood and the cortex stores the urine
- D. The glomerulus filters the blood and the tubule reabsorbs useful substances such as water**

Correct answer: D. Filtration happens at the glomerulus and Bowman's capsule in the cortex. Useful substances and water are then reabsorbed along the tubule, including the loop of Henle. The pelvis only collects urine.

USE AFTER: OBSERVE PHASE — Observe: evidence table (Animal | Environmental challenge | Waste product | Organ or behaviour | How it keeps balance)

Q3. Which row correctly matches the animal to its main nitrogenous waste?

- A. Freshwater fish: uric acid; mammals: ammonia; insects: urea
- B. Freshwater fish: ammonia; mammals: uric acid; insects and birds: urea
- C. Freshwater fish: ammonia; mammals: urea; insects and birds: uric acid**
- D. Freshwater fish: urea; mammals: uric acid; birds: ammonia

Correct answer: C. Freshwater fish excrete ammonia, which is very toxic but easily diluted in plenty of water. Mammals excrete urea, and insects and birds excrete uric acid, which saves water.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning explanation for the four animals (freshwater fish)

Q4. A tilapia from Lake Victoria produces large amounts of dilute urine. What is the best explanation?

- A. It makes dilute urine because it does not need to excrete nitrogenous wastes
- B. Water enters its body by osmosis, so its kidneys remove the excess water as dilute urine**
- C. Its body is saltier than the lake water, so it loses water by osmosis and must make dilute urine
- D. It drinks a lot of lake water and must remove the extra salt in urine

Correct answer: B. A freshwater fish has body fluids with more dissolved substances than the surrounding water, so water enters by osmosis. The kidney removes the excess water as large volumes of dilute urine, which also carries away wastes.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning explanation (panting dog and basking lizard)

Q5. On a hot afternoon a dog pants with its tongue hanging out, while a lizard sits still on a warm rock. Which statement is correct?

- A. Panting and basking both make the animals produce more urine to lose heat
- B. Panting warms the dog's body, and the lizard stays still because it makes its own body heat
- C. Panting mainly removes nitrogenous waste from the dog, and the lizard cannot control its temperature
- D. Panting increases evaporation to cool the dog, and the lizard controls its temperature by behaviour such as basking or moving to shade**

Correct answer: D. The dog is an endotherm and loses heat as water evaporates from its tongue and mouth. The lizard is an ectotherm that regulates its temperature by behaviour, basking to warm up and moving to shade when too hot.

USE AFTER: EXPLAIN PHASE — Explain: concept map or flowchart of a negative feedback loop (blood sugar regulation)

Q6. After a meal rich in sugar, a person's blood glucose rises above the set point. Which sequence shows negative feedback?

- A. Insulin is released, blood glucose rises further, and the pancreas stops working
- B. Glucagon is released, the liver releases more glucose, and blood glucose rises even further
- C. Insulin is released, cells and the liver take up glucose, and blood glucose falls back towards the set point**
- D. The kidney makes ADH, and blood glucose returns to the set point

Correct answer: C. In negative feedback the response reverses the change. High blood glucose triggers insulin, which lowers glucose back towards the set point.

USE AFTER: DQB PHASE — DQB wall: 'Why do diabetics pass a lot of urine?' (diabetes insipidus and ADH)

Q7. A patient passes very large volumes of dilute urine because the body does not make or respond to enough ADH. What explains this?

- A. The liver produces too much urea, which pulls water into the urine
- B. The skin reabsorbs water and stops sweating
- C. The glomerulus filters too little blood
- D. The kidney tubules reabsorb too little water, so more water is lost in urine**

Correct answer: D. ADH makes the kidney tubules and collecting ducts reabsorb water. Without enough ADH, little water is reabsorbed and a large volume of dilute urine is passed. This is diabetes insipidus.

USE AFTER: MODEL BUILDING PHASE — Model: Final Model (organs for excretion and wastes: liver deamination and urea formation)

Q8. When learners add the liver to their final model to show where wastes come from, which statement is correct?

- A. The liver makes ADH, which turns ammonia into uric acid
- B. The liver filters blood in nephrons and forms urine
- C. The liver makes sweat, which carries urea out through the skin only
- D. The liver removes the amino group from excess amino acids (deamination) and forms urea, which the kidneys excrete**

Correct answer: D. Deamination in the liver removes nitrogen from excess amino acids and converts it to urea. The urea is carried in the blood to the kidneys and is excreted in urine. Some also leaves in sweat.

USE AFTER: END OF LESSON — End of lesson: exit ticket and summary table (why excretion and homeostasis matter)

Q9. Which statement best explains why excretion and homeostasis are important for the survival of body cells?

- A. Excretion adds water to cells, and homeostasis makes the external environment stable
- B. Excretion and homeostasis are only needed when an animal is ill or in a hot place
- C. Excretion keeps the body temperature steady, and homeostasis removes urea from the blood
- D. Excretion removes toxic metabolic wastes from cells, and homeostasis keeps conditions such as temperature and water level steady**

Correct answer: D. Metabolic wastes such as ammonia and urea are toxic if they build up. Homeostasis keeps the internal conditions steady so that cells can work and stay alive, and when either fails health suffers.